Physica Scripta

PAPER

Theoretical predictions of nuclear binding energy for the observed nuclei: the influence of coefficients and terms in a semi-empirical mass formula

To cite this article: A F Saad and Zeyad I Elghobary 2024 *Phys. Scr.* **99** 085308

View the article online for updates and enhancements.

You may also like

- Basic characteristics of nuclear landscape by improved Weizsäcker-Skyrme-type nuclear mass model
Na-Na Ma, , Hai-Fei Zhang et al.

- Improvement of nuclear semi-empirical mass formula by including shell effect
Qing Wu, , Wei-Feng Li et al.

- Mass predictions of exotic nuclei within a macroscopic-microscopic model
G Royer, A Onillon, M Guilbaud et al.

This content was downloaded from IP address 159.149.167.161 on 28/05/2026 at 14:05

Physica Scripta

RECEIVED
29 November 2023

REVISED
16 June 2024

ACCEPTED FOR PUBLICATION
10 July 2024

PUBLISHED
25 July 2024

PAPER

Theoretical predictions of nuclear binding energy for the observed nuclei: the influence of coefficients and terms in a semi-empirical mass formula

A F Saad and Zeyad I Elghobary

Physics Department, Faculty of Science, Zagazig University, PO 44519, Zagazig, Egypt

E-mail: abdallah.saad2256@gmail.com, abdallahsaad56@hotmail.com and afsaad@zu.edu.e.g

Keywords: semi-empirical mass formula (SEMF), liquid drop model (LDM), nuclear binding energy, nuclear masses, coefficients, the experimentally measured nuclei AME2020

Abstract

The aim of this study is to calculate the nuclear binding energy for the 2457 nuclei which experimentally observed, as tabulated in the atomic mass evaluation AME2020, by using the semi-empirical mass formula based on the renowned liquid drop model (LDM). The Bethe-Weizsäcker (BW) mass formula-four terms (BWMF-4T) is extended to incorporate an additional six terms, resulting in a formula of ten terms. Our findings demonstrate commendable accuracy and reliability, as substantiated by the acceptable level of uncertainty observed when compared them to experimental data and existing models. The influence of coefficients and terms in the semi-empirical mass formula has been determined. All possible combinations of the terms are fitted in turn to the measured nuclear masses, and the outcomes are analyzed in order to reveal correlations and mutual influences between the various terms in all observed mass regions of the periodic table. The fitted surface energy and symmetry coefficients in the semi-empirical mass formula-ten terms (SEMF-10T) are remaining 26.4 and 33.3 MeV respectively. In contrast, coefficients fitted in the classic BW formula with and without one or two additional terms are around 17 and 23 MeV, respectively. The interplay between different terms in this mass formula is found to be significant and interesting.

1. Introduction

The Weizsäcker mass formula, also known as the Bethe-Weizsäcker (BW) mass formula, has been derived from fundamental principles [1, 2]. but the roots of this formula dates back to 1930 by G. Gamow [3]. Indeed, it should be noted that this formula has been developed to align with previous experimental and theoretical findings while satisfying specific principles. We aim to determine new coefficients for the BW mass formula, a foundational equation utilized for calculating the nuclear masses of nuclei far from stability and stable nuclides of the periodic table. By following this approach, we contribute to refining the BW mass formula and improving its ability to predict nuclear masses and describe the decay behavior of these nuclides. The determination of nuclear mass holds paramount importance in the field of nuclear physics, encompassing a wide range of fundamental properties. Within this context, the investigation of superheavy elements (SHE) and heavy elements (HE) stands as a highly significant sub-field within the realm of low-energy nuclear physics [4]. While the prospect of observing elements proximate to $Z\sim120$ and $Z = 122$ appears promising, the same cannot be affirmed for elements with $Z = 126$, as their near-future observability appears to be remote.

The BW mass formula calculates the binding energy of a nucleus in its ground state. Experimental observations indicate that nuclei characterized by double magic (N) and (Z) values, or at least one of them, exhibit significantly elevated binding energy levels when compared to the smoothly varying theoretical predictions derived from the semi-empirical mass formula. This foundational property of nuclei holds profound significance not only within the domain of nuclear physics [5] but also in the realm of nuclear astrophysics [6, 7]. The determination of nuclear masses for nuclei far from stability serves as a key assessment of

© 2024 IOP Publishing Ltd
All rights, including for text and data mining, AI training, and similar technologies, are reserved.

our understanding of nuclear structure. In recent decades, there has been significant progress in the development of new theoretical frameworks aimed at accurately determining the masses of nuclei that are far from stability. Divergence among nuclear models becomes increasingly pronounced when moving away from β stability, leading to notable disparities in predictions. Consequently, the determination of nuclear masses remains a significant challenge, both in terms of experimental measurements and theoretical understanding.

Merely 2457 out of the numerous nuclides known to populate the nuclear landscape exhibit stability or near-stability, collectively forming what is commonly referred to as the 'valley of stability' [8]. Introducing additional nucleons leads nuclides to deviate from this stable region, leading us into the vast expanse of short- or long-lived radioactive nuclei. These nuclei undergo disintegration via the emission of β- and α-particles or both, or alternatively, may undergo spontaneous fission, in the case of heavy and superheavy nuclides. The experimental range of the nuclear landscape continues to evolve each year, with the addition of newly rare isotopes on an annual basis. In the present context, it is important to note that the theoretical range of the chart of nuclides is subject to constant evolution. Therefore, any predictions made should be seen as contingent upon critical analysis and potential adjustments.

This work aims to determine the coefficients of different combinations of terms of the classic Bethe-Weizsäcker (BW) mass formula, including the pairing term and in the BW mass formula with five additional terms. This determination is carried out through a least square fitting procedure to the experimentally available atomic masses AME2020 [9]. Additionally, the study seeks to achieve a concise comprehension of the correlation between nuclear masses and the neutron number (N) and proton number (Z), employing theoretical reasoning to establish the functional form of the N-Z dependence. Subsequently, the experimentally measured masses are utilized to ascertain the coefficients associated with the different terms of the proposed mass formula. This formulated approach, incorporating the newly determined coefficients, is then employed to predict the nuclear binding energies of both stable nuclei and nuclei situated far from the stability line, with a specific focus on all observed nuclei tabulated in the atomic mass evaluation AME2020.

2. The Bethe–Weizsäcker (BW) mass formula (BWMF) with additional terms

The semi-empirical mass formula, originally devised by cf von Weizsäcker [1], is alternatively known as the Weizsäcker formula or the Bethe–Weizsäcker mass formula [2]. This formula is widely utilized for estimating atomic nucleus mass and the prediction of diverse decay modes contingent based on the nucleus's proton and neutron counts. In its early stages, George Gamow envisioned the nucleus as analogous to a water droplet, with particles held together by surface tension, a perspective that later gave rise to the liquid drop model [8]. Gamow's contributions included a significant portion of the terms within the formula, providing initial estimations for coefficient values, while cf von Weizsäcker further developed this formula [1]. Throughout the years, refinements have been made to fine-tune these coefficients. However, the fundamental structure of the formula has remained largely unchanged, with incorporation of some additional terms over time.

The coefficients representing the various terms in the mass formula, which are determined through the least square procedure, are provided along with their standard errors, with emphasis on the least significant digits of the coefficients. The four fundamental terms of the Bethe–Weizsäcker (BW) mass formula are derived from equation (1) [1].

$$B(Z, N) = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{sy} \frac{(Z - N)^2}{A} \tag{1}$$

Previous studies have shown that a set of four basic terms is sufficient to describe the essential features of potential barriers associated with fission, fusion, cluster, and alpha decay. These barriers are primarily influenced by the nature of nuclear binding energy evolution [10–16]. However, there remains an urgent need for refinements for enhancements in the volume, surface, Coulomb, and nuclear proximity energy terms to align more closely with experimental outcomes. Furthermore, the inclusion of a pairing energy term has been found crucial for elucidating structural phenomena and improving quantitatively the results. Consequently, an additional fifth term in the form of a pairing term was introduced [17, 18] and resulting in the following representation of equation (1) to be as a BW plus pairing term (PT) or as we named (BW + PT) mass formula:

$$B(Z, N) = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{sy} \frac{(Z - N)^2}{A} \pm a_p \delta(Z, N) \tag{2}$$

As $\delta(Z, N)$ assumes a negative value for odd Z, odd N nuclei, positive for even Z, even N nuclei and vanishes for odd A nuclei. Equation (2) can be rearranged as follows [19]:

Table 1. Best fit values of coefficients of the semi-empirical mass formula based on experimental data for 2457 nuclei.

BWMF plus extra terms	a_v	a_s	a_C	a_{sy}	a_p	a_w	a_{xC}	a_{st}	a_R	Shell term	
										a_m	β_m
BW	15.52552	−16.89493	−0.70218	−22.98742	–	–	–	–	–	–	–
BW + PT	15.54143	−16.94442	−0.70330	−23.02135	12.49976	–	–	–	–	–	–
BW + PT + Wigner	15.61154	−17.27070	−0.70552	−23.38225	12.48607	12.27334	–	–	–	–	–
BW + PT + Coulomb exch.	15.40844	−17.36595	−0.71075	−22.17975	12.19299	–	0.71964	–	–	–	–
BW + PT + Surface sym.	15.65047	−17.51526	−0.70307	−26.50109	12.53725	–	–	18.41388	–	–	–
BW + PT + Shell	15.60474	−17.04580	−0.70615	−23.23764	12.06599	–	–	–	–	−1.000357	0.04675
BW + PT + Wigner + Shell	15.68279	−17.41108	−0.70850	−23.64392	12.08422	13.79401	–	–	–	−0.99956	0.04596
BW + PT + 5T	16.58173	−26.36067	−0.76016	−33.33164	11.43385	−62.50588	1.76145	64.74710	14.36290	−1.09613	0.04881

a_v −Volume term coefficient, a_s−Surface term coefficient, a_C−Coulomb term coefficient, a_{sy} −Symmetry term coefficient, a_p −Pairing term coefficient, a_W− Wigner term coefficient, a_{xC}− Coulomb Exchange term coefficient, a_{st}−Surface-symmetry term coefficient, a_R−Curvature term coefficient and (a_m , β_m)−Shell term coefficients.

$$B(Z,N) = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{sy} \frac{(Z-N)^2}{A} + a_p \frac{(-1)^Z + (-1)^N}{2A^{1/2}} \tag{3}$$

This formula provides the baseline fit relative to which all refinements can be refereed. The fit to the 2457-mass table from the AME2020 mass produces constant coefficient values, as shown in table 1. To further refine the model, the classic (BW plus PT) mass formula is extended to include an extra five terms as follow:

The BW plus PT mass formula and the Wigner term with a coefficient (a_W):

$$B(Z,N) = a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) + a_w \frac{|N-Z|}{A} \tag{4}$$

The BW plus PT mass formula and the Coulomb Exchange term with coefficient (a_{xc}):

$$B(Z,N) = a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) + a_{xc} \frac{Z^{4/3}}{A^{1/3}} \tag{5}$$

The BW plus PT mass formula and the surface symmetry term with coefficient (a_{st}):

$$B(Z,N) = a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) + a_{st} \frac{(N-Z)^2}{A^{4/3}} \tag{6}$$

The BW plus PT mass formula and the shell term with coefficient (a_m, β_m):

$$B(Z,N) = a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) + a_m P + \beta_m P^2 \tag{7}$$

The BW plus PT mass formula as well as two terms of Wigner and shell:

$$B(Z,N) = a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) + a_w \frac{|N-Z|}{A} + a_m P + \beta_m P^2 \tag{8}$$

Finally, the classic Bethe–von Weizsäcker (BW) semi-empirical mass formula plus pairing term has been expanded with five additional terms a it can be written as follows:

$$\begin{aligned} B(Z,N) &= a_v A + a_s A^{2/3} + a_c \frac{(Z)^2}{A^{1/3}} + a_{sy} \frac{(N-Z)^2}{A} \pm a_p \delta(Z,N) \\ &+ a_w \frac{|N-Z|}{A} + a_{xc} \frac{Z^{4/3}}{A^{1/3}} + a_{st} \frac{(N-Z)^2}{A^{4/3}} + a_R A^{1/3} + a_m P + \beta_m P^2 . \end{aligned} \tag{9}$$

In the classic BW mass formula, the volume energy corresponds to the saturated exchange force and infinite nuclear force matter with a short-range nature of these nuclear forces. The second term, the surface energy term, highlights the reduced binding energy of the nucleons at or near the surface of the nucleus. Due to this reduced binding energy, the contribution of surface nucleons to the total binding energy differs from the full saturation binding assumed to be uniform for all nucleons. The third term, the Coulomb energy, accounts for the loss of binding energy as a result of the repulsion between the intranuclear protons. The last term in this classic BW formula is the symmetry term which reduces the binding energy for a given A nucleus in the case of the proton and neutron numbers are not equal. Thus, the reduction in the binding energy arises from the asymmetry between neutrons and protons. The pairing term is a purely empirical term based on the experimental values of nuclear binding energies for even–even or odd–odd nuclei. The pairing term has been calculated as previously mentioned in the equations (2) and (3). This term is necessary to further improve the agreement with the experimental values of the binding energies. The first extra term is the Wigner term which appears in the counting of identical pairs, p-p or n-n pairs, in a nucleus based on experimental evidence. The second extra term, the Coulomb exchange term, is expected as a quantal correction to the classical Coulomb energy formula. The third extra term is the surface symmetry term, which appears to include a surface correction to the volume energy but not to the symmetry energy. The fourth extra term is the curvature energy term, which seems as a correction to the surface energy when the surface energy is considered as a function of local properties of the surface and accordingly depends on the mean local curvature. The last extra term is a shell correction which may be parametrized [18 and reference therein] as stated in equation (9). This treatment of these shell effects can be based on Casten's observation [17] that widely scattered nuclear data tend to cluster significantly when plotted against the valence-nucleon variable:

$$P = \frac{\upsilon_n \upsilon_p}{\upsilon_n + \upsilon_p} \tag{10}$$

where υ_n and υ_p are the numbers of valence nucleons (the difference between the actual nucleon numbers, N and Z, respectively, and the nearest magic numbers). It should be noted that although it is referred to this term as a single term, the shell correction involves two fit coefficients.

The discrepancies in nuclear binding energy between the theoretical predictions and experimental data are defined as follows:

Table 2. Comparison of the coefficient values presented in the current work to those of previous literature studies.

	a_v	a_s	a_C	a_{sy}	a_p	a_w	a_{xC}	a_{st}	a_R	Shell Term	
										a_m	β_m
Present Work (BW + PT + 5T)	16.5817	−26.3607	−0.7602	−33.3316	11.4339	−62.5059	1.7615	64.7471	14.3629	−1.0961	0.0488
Present Work (BW + PT)	15.5414	−16.9444	−0.7033	−23.0214	12.4998	–	–	–	–	–	–
[16][a]	15.7827	−17.9042	−0.72404	−23.7193	–	–	–	–	–	–	–
[18]	16.5800	−26.9500	−0.7740	−31.5100	9.8700	−43.4000	2.2200	55.6200	14.7700	−1.9000	0.14000
[20]	15.2580	−16.2600	−0.6890	−22.2000	10.0800	–	–	–	–	–	–
[21]	15.7770	−18.3400	−0.7100	−23.2100	12.0000	–	–	–	–	–	–
[22]	19.1223	−18.1929	−0.5187	−12.5405	–	28.9866	–	–	–	–	–
[23]	16.2400	−18.6300	–	−32.6500	–	–	–	–	–	–	–
[24]	14.9297	−15.058	−0.6615	−21.6091	10.1744	–	–	–	–	–	–

[a] The coefficient values for the volume term are ranging from 14.8497 to 15.7335 and from 15.9622 to 15.3737 and the coefficient values for the surface term are ranging from 11.2996 to 18.0625 and from 14.9364 to 18.194 as tabulated in Table I and Table II, respectively. The Coulomb term depends on the nuclei radius with $R_0 = 1.28A^{1/3} - 0.76 + 0.8A^{-1/3}$, the coefficient values of the Wigner term a_w are ranging from 26.4 to 44.2 and from 27.8 to 44.4 as tabulated in Table I and Table II, respectively.

Table 3. Root mean square deviation (σ) values in MeV for predicted 2457 nuclei's disparity $|\Delta B(Z,\ N)|$ based on best fit values calculations.

BW mass formula with extra terms	σ (MeV)	Minimum $|\Delta B(Z,\ N)|$ (MeV)	Maximum $|\Delta B(Z,\ N)|$ (MeV)
BW	3.19441	0.00039	17.12990
BW + PT	3.06667	0.00168	14.85390
BW + PT + Wigner	3.03500	0.00088	14.84330
BW + PT + Coulomb exch.	3.01683	0.00241	15.10880
BW + PT + Surface sym.	2.80814	0.00016	14.89880
BW + PT + Shell	2.76130	0.00159	15.07600
BW + PT + Wigner + Shell	2.71697	0.00091	12.55690
BW + PT + 5T	1.88618	0.00079	7.91555

$$\Delta B(Z,\ N) = B^{th}(Z,\ N) - B^{\exp}(Z,\ N) \tag{11}$$

$\Delta B(Z,\ N)$ is the variation of nuclear binding energy in MeV between the theoretical predictions and experimental data for the semi-empirical mass formula. The root mean square deviation (σ) is defined by the following expression:

$$\sigma = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n}(B_i^{th}(Z,\ N) - B_i^{\exp}(Z,\ N))^2} \tag{12}$$

Where n is the total number of nuclei. ΔB is the deviation of theoretical predictions from the experimental values. This root mean square deviation (σ) has been used to measure the accuracy of the results.

3. Results and discussion

In the current study, we use the classic Bethe-Weizsäcker semi-empirical mass formula both with and without additional terms, to examine the influence of coefficients and terms effects on the nuclear binding energy for the 2457 observed nuclei as tabulated in the atomic mass evaluation AME2020 [9]. The best-fit values of coefficients for the semi-empirical mass formulae based on experimental data for 2457 nuclei are listed in table 1. Based on this fit, the coefficients of the basic four terms of the full BW mass formula (BWMF-10T) are much higher than those of the classic BW formula as shown in table 1. Specifically, the value of the pairing coefficient a_p is around 12 in the BW formula with and without one or two additional terms, whereas in the case of a full mass formula (BWMF-10T), it is around 11, as seen in table 1. The estimates of a_{xC} are clearly a little bit more well-determined than a_C as given by the best fit to data of the BW mass formula, whereas in the case of the full mass formula of equation (9), which is displayed in table 1. The magnitude of this coefficient exhibits a significant and consistent increase for more than twice the value of the coefficient a_C of the BW mass formulae and a_{xC} of the BW formula with two extra terms: pairing and Coulomb exchange terms. In the case of adding the surface symmetry term to the BW formula plus the pairing term, the coefficient of the asymmetry energy term increased by approximately 15.2%. Moreover, the surface symmetry term of the BW formula-ten terms drastically increased by a factor of approximately three and a half times compared to that of the BW formula with the two extra terms: paring and surface symmetry terms. Conversely, upon the addition of the Wigner term, the coefficient (a_W) drastically changed with the opposite sign from 12.27334 MeV of the BW formula-six terms to −62.50588 compared to that of the full BW formula-ten terms. On the contrary, the coefficient of two shell terms a_m and β_m is a little bit increased in the case of the full BW formula-ten terms compared to the BW formula plus paring and shell terms. A comparison of the coefficient values presented in the current work to those of previous studies is considered as shown in table 2. The values of the coefficients associated with the basic four terms, volume, surface, Coulomb, and asymmetry, in addition to the curvature term, which correspond to values equal 16.5817, −26.3607, −0.7602, −33.3316 and 14.3629 MeV, agree excellently with the theoretical values 16.5800, −26.9500, −0.7740, −31.5100 and 14.7700 MeV, respectively, produced by Kirson [18] which confirms the validity of the model used, whereas the coefficients of other extra terms are in reasonable agreement. There is also a good agreement in the case of the BW mass formula-five terms with the coefficient of the volume term of references [20, 21], except for references Mavrodiev and Deliyergiyev [22], Myers and Swiatecki [23] and Benzaid *et al* [24] listed in table 2, the first reference reports significantly higher values, while the second reports values that are a little bit higher and the third reports values are smaller than the expected value. Analogous behavior was noted for the coefficients of the other terms, as can be seen in table 2. On the other hand, Royer and Gautier [16] determined the coefficients of different combinations of terms of the semi-empirical mass formula based on the liquid drop model as seen in table 2. They studied the dependence of the energy coefficient values (in MeV) on the selected term set including or not the pairing and theoretical shell energies which gives significant improvement of the experimental data reproduction when additional terms are added to the three basic,

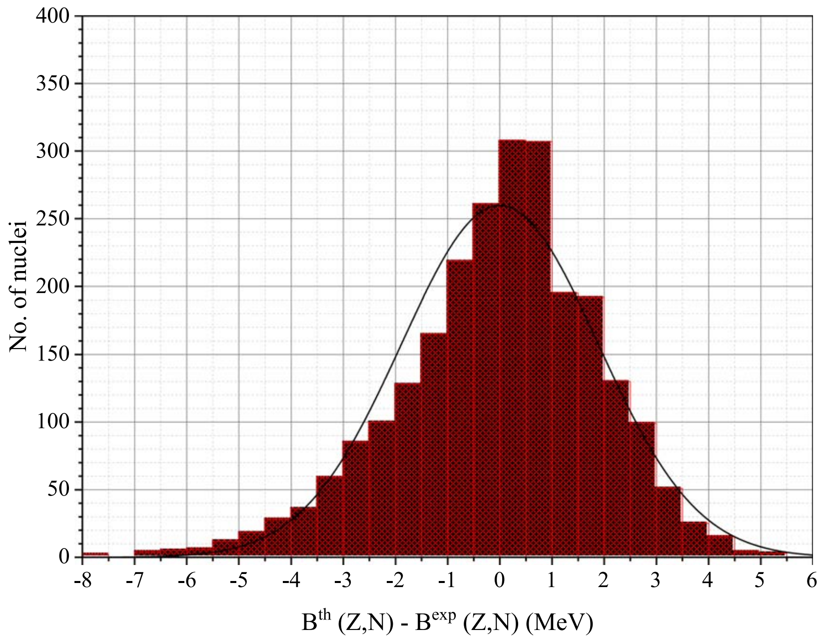

Figure 1. Shows the histogram of the deviations of the calculated binding energies from the experimental data versus the number of nuclei. There are 2457 isotopes known experimentally as of AME2020.

Table 4. Distribution of nuclei through absolute value of disparity $|\Delta B(Z,\ N)|$ and root mean square deviation (σ).

$\|\Delta B(Z,\ N)\|$ MeV	No. of nuclei	σ (MeV)
0.001 → 1	1095	0.56186
0.001 → 2	1775	1.03021
0.001 → 3	2189	1.41559
0.001 → 4	2360	1.64394
0.001 → 7	2457	1.88618
4 → 7	97	4.93588

Table 5. Comparison of root mean square deviation (σ) between present work and different models.

Model	σ	References
Present Work (BW + PT + 5T)	1.643	–
LDM	1.170	[16]
RMF	2.217	[25]
HFB-21	0.572	[26]
DZ31	0.397	[27]
ETFSI-2	0.719	[28]
KTUY	0.701	[29]
FRDM	0.661	[30]
WS3	0.335	[31]

volume, surface and Coulomb energy terms. Consequently, the difference ΔE (in MeV) between the theoretical and experimental masses for the 1522 nuclei evaluated by Royer and Gautier [16] does not exceeds 2.25 MeV and the root mean square deviation $\sigma = 1.17$ MeV which is a great success for their modifications of the liquid drop model coefficients and terms. However, the current work for the 1775 nuclei (in our case, ΔE does not exceeds 2.0 MeV and the root mean square deviation $\sigma = 1.0302$ MeV) as seen in table 4 which is in excellent agreement with the work done by Royer and Gautier [16], beyond this limit the agreement is fairly lesser thereafter.

The minimum, maximum, and root mean square deviation of the absolute value $|\Delta B(Z,\ N)|$ are shown in table 3. The fit of the BW mass formula with no extra terms produces a root mean square deviation $\sigma =\ 3.19441$ MeV, while the fit of the BW mass formula with pairing term gives $\sigma =\ 3.06667$ MeV. Thus, the effect of the

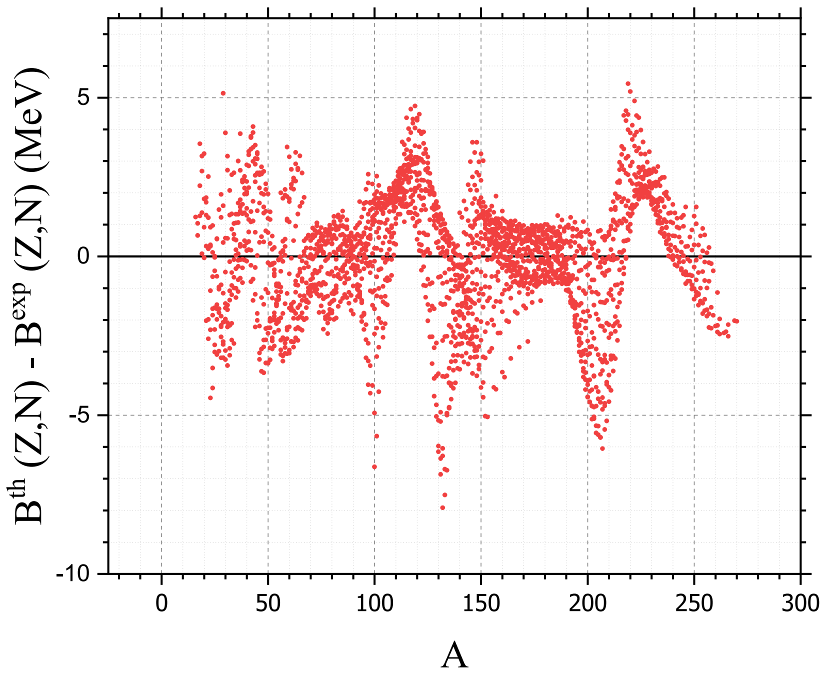

Figure 2. Difference (in MeV) between the theoretical and experimental binding energies for the 2457 nuclei as a function of the mass number A.

pairing term is very small. It should be noted that the successive inclusion of the extra five terms: Wigner, Coulomb exchange, surface symmetry, and shell, as well as both two terms added together, shell and Wigner, which systematically reduces the root mean square deviation σ from 3.06667 to 3.03500, 2.80818, 2.76130, 2.71697, and 1.88618 MeV, respectively. The pairing term and the Wigner term taken alone do not allow for a better fit to the experimental binding energies while the surface symmetry and the shell term separately have a strong effect on decreasing σ. When both shell and Wigner contributions to the binding energy are taken together into account $\sigma = 2.71697$ MeV which is a satisfactory value. The addition of the Coulomb exchange and Wigner energy terms to the pairing term, taken separately, does not result in significantly improve rms deviation, as shown in table 3. Finally, it's important to highlight that inclusion of the pairing term plus five extra terms significantly reduce the root mean square deviation by less than a factor of two approximately.

On the other hand, when and only when this fit is applied to a lesser number of nuclei 2360 of the nuclear landscape AME2020 it reveals a significant improvement in the root mean square deviation $\sigma = 1.64394$ MeV, However, the excluded 97 nuclides exhibit an extreme discrepancy between theoretical and experimental values of binding energy $|\Delta B(Z, N)|$ ranging from 4 to 7 MeV, as seen in figure 1. Their rms deviation is $\sigma = 4.93588$ which indicating a high error as evident in table 4. In the case of the absolute value $|\Delta B(Z, N)|$ ranging from 0.001 to 1.0 MeV, the 1095 nuclides have a rms deviation of $\sigma = 0.56186$ MeV, which represents a reduction in the rms deviation by more than a factor of five — a very satisfactory result [18]. Conversely, for the 1775 nuclides, the difference between the theoretical and experimental binding energies never exceeds 2 MeV, resulting in a corresponding root mean square deviation of $\sigma = 1.03021$. In the same context, for the 2189 nuclides, $\Delta B(Z, N)$ ranges from 0.001 to 3 MeV, with a rms deviation of $\sigma = 1.41559$. For a larger number of nuclei 2360, the variation of $\Delta B(Z, N)$ does not exceed 4 MeV, with a rms deviation of $\sigma = 1.64394$, which is in excellent agreement with the previous study [18].

Figure 2 shows the variation of nuclear binding energy between experimental and theoretical predictions values using BWMF with six additional terms for the 2457 observed nuclei, showing the $\Delta B(Z, N)$, in (MeV), as a function of mass number (A). This difference between theoretical and experimental binding energies does not exceed 4 MeV for 2360 nuclides and is less than 1 MeV for 1095 nuclei, while 97 nuclides only have a higher variation up to 7 MeV, as shown in table 4 and mentioned above in more details.

In figures 3(a)–(h), we show the deviations of the calculated binding energies from the experimental data of the BW mass formula with and without extra terms. For all the 2457 nuclei with Z and $N \geqslant 8$, the deviations are ranging from 0.0001 to around 4 MeV for the 2360 nuclei. These deviations illustrate the difference between the measured binding energies with the full BW-ten terms (BWMF-10 terms) and those with the BW plus one or two terms. A useful comparison of root mean-square deviation σ between the present work and other different

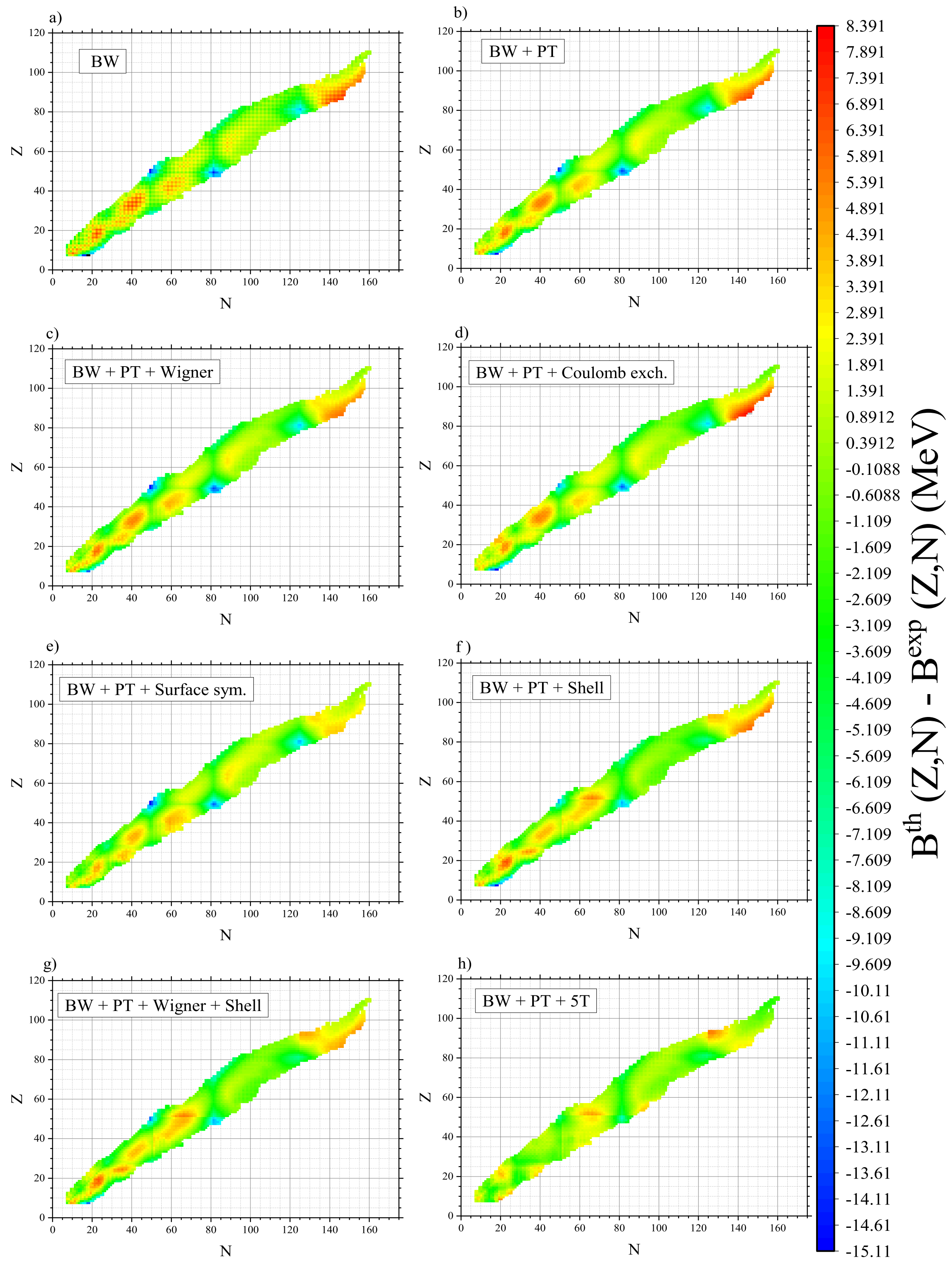

Figure 3. Map of stable and unstable nuclei as a function of Z and N. Color on line sees the difference between measured and calculated binding energies with the (a) BW formula, (b) BW with pairing term, (c) BW with pairing and Wigner terms, (d) BW with pairing and Coulomb exchange terms, (e) BW with pairing and surface symmetry terms, (f) BW with pairing and shell terms, (g) BW with pairing, shell and Wigner terms and (h) BW with paring term and 5 extra terms (full BW-ten terms).

models [25–31] as seen in table 5. For most nuclei, the results of BWMF-10 terms are consistent in general (with deviations σ smaller than two MeV).

4. Conclusion

The binding energy for stable and unstable nuclei has been studied using the basic four terms of the BW mass formula with and without six additional terms. These six extra terms: pairing, Wigner, Coulomb exchange, surface symmetry, curvature, and shell correction, can be added individually or in various combinations till reaches ten-term formula, to test the efficiency of each term and the method the different terms interact with one another. The binding energy for the 2457 nuclides was estimated, and theoretical binding energies for 2360 nuclei from the nuclear chart, utilizing the newly determined coefficient parameters of the semi-empirical mass formula (SEMF–10T), demonstrate good agreement with experimental data and with some other models. One of the most significant results is that the surface energy coefficient remains around 26 MeV, while the surface-asymmetry coefficient ranges from 18.4 to 64.7 MeV with a positive sign, as well as the Wigner term coefficient is ranging from 12.3 to −62.5 MeV with opposite sign. Additionally, the accuracy of the fitting strongly depends on the selected shell and pairing energies, as well as on the surface symmetry and Wigner energy terms. In conclusion, there are significant correlations between all the different terms of the mass formula, highlighting the intricate interplay within the nuclear binding energy.

Acknowledgments

We wish to give our sincere gratitude to Prof M Taha the Head of the Mathematics and Theoretical Physics Department, Nuclear Research Centre, Egyptian Atomic Energy Authority for devoting his invaluable time to give us comprehensive comments which significantly improved our work.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

ORCID iDs

A F Saad https://orcid.org/0000-0001-9629-4581
Zeyad I Elghobary https://orcid.org/0009-0002-0230-9047

References

[1] Weizsäcker C V 1935 Zur theorie der kernmassen *Z. Phys.* **96** 431–58
[2] Bethe H A and Bacher R F 1936 Nuclear physics. A. Stationary states of nuclei *Rev. Mod. Phys.* **8** 82–229
[3] Afanasjev A V, Agbemava S E and Gyawali A 2018 Hyperheavy nuclei: existence and stability *Phys. Lett.* B **782** 533–40
[4] Lunney D, Pearson J M and Thibault C 2003 Recent trends in the determination of nuclear masses *Rev. Mod. Phys.* **75** 1021
[5] Mumpower M R, Surman R, Fang D L, Beard M, Möller P, Kawano T and Aprahamian A 2015 Impact of individual nuclear masses on r-process abundances *Physical Review* C **92** 035807
[6] Schatz H and Ong W J 2017 Dependence of x-ray burst models on nuclear masses *Astrophys. J.* **844** 139
[7] Erler J, Birge N, Kortelainen M, Nazarewicz W, Olsen E, Perhac A M and Stoitsov M 2012 The limits of the nuclear landscape *Nature* **486** 509–12
[8] Gamow G 1930 Mass defect curve and nuclear constitution *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* **126** 632–44
[9] Wang M *et al* 2021 *Chinese Phys.* C **45** 030003
[10] Royer G and Remaud B 1984 Fission processes through compact and creviced shapes *J. Phys. G: Nucl. Phys.* **10** 1057
[11] Royer G and Remaud B 1985 Static and dynamic fusion barriers in heavy-ion reactions *Nucl. Phys.* A **444** 477–97
[12] Royer G 2000 Alpha emission and spontaneous fission through quasi-molecular shapes *J. Phys. G: Nucl. Part. Phys.* **26** 1149
[13] Royer G and Moustabchir R 2001 Light nucleus emission within a generalized liquid-drop model and quasimolecular shapes *Nucl. Phys.* A **683** 182–206
[14] Royer G and Zbiri K 2002 Asymmetric fission for 70, 76Se and 90, 94, 98Mo via quasimolecular shapes and related formulas *Nucl. Phys.* A **697** 630–8
[15] Gherghescu R A and Royer G 2003 Shape isomerism of rotating 44 Ti and 48 Cr *Phys. Rev.* C **68** 014315
[16] Royer G and Gautier C 2006 Coefficients and terms of the liquid drop model and mass formula *Phys. Rev.* C **73** 067302
[17] Casten R 2000 *Nuclear Structure from a Simple Perspective* vol 23 (Oxford University Press)
[18] Kirson M W 2008 Mutual influence of terms in a semi-empirical mass formula *Nucl. Phys.* A **798** 29–60
[19] Belyaev A and Ross D A 2021 *The Basics of Nuclear and Particle Physics.* (Springer) 37–52
[20] Chowdhury P R, Samanta C and Basu D N 2005 Modified bethe–weizsäcker mass formula with isotonic shift and new driplines *Mod. Phys. Lett.* A **20** 1605–18
[21] Samanta C and Adhikari S 2004 Shell effect in Pb isotopes near the proton drip line *Nuclear Physics*A **738** 491–4
[22] Mavrodiev S C and Deliyergiyev M A 2018 Modification of the nuclear landscape in the inverse problem framework using the generalized Bethe–Weizsäcker mass formula *Int. J. Mod. Phys.* E **27** 1850015
[23] Myers W D and Swiatecki W J 1996 Nuclear properties according to the Thomas-Fermi model *Nucl. Phys.* A **601** 141–67
[24] Benzaid D, Bentridi S, Kerraci A and Amrani N 2020 Bethe–Weizsäcker semiempirical mass formula coefficients 2019 update based on AME2016 *Nucl. Sci. Tech.* **31** 9
[25] Geng L, Toki H and Meng J 2005 Masses, deformations and charge radii—nuclear ground-state properties in the relativistic mean field model *Prog. Theor. Phys.* **113** 785–800

[26] Goriely S, Chamel N and Pearson J M 2010 Further explorations of Skyrme-Hartree–Fock-Bogoliubov mass formulas. XII. Stiffness and stability of neutron-star matter *Phys. Rev.* C **82** 035804

[27] Zuker A P 2008 Shell formation and nuclear masses *Revista Mexicana De Física* **54** 129–35

[28] Goriely S 2000 Tenth Int. Symp. on capture gamma-ray spectroscopy and related topics ed S Wender *AIP Conf. Proc.* 529, 287

[29] Koura H, Tachibana T, Uno M and Yamada M 2005 Nuclidic mass formula on a spherical basis with an improved even–odd term *Prog. Theor. Phys.* **113** 305–25

[30] Möller P, Sierk A J, Ichikawa T and Sagawa H 2016 Nuclear ground-state masses and deformations: FRDM (2012) *At. Data Nucl. Data Tables* **109** 1–204

[31] Wang N and Liu M 2011 Nuclear mass predictions with a radial basis function approach *Phys. Rev.* C **84** 051303